Reproducing and Analysing Training SVMs Without Offset

AML Mid-Semester Examination — Part B Report

Kush Agarwal Roll No. 230050
NST, Rishihood University, Sonipat

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Abstract

This report documents the reproduction, ablation, and failure-mode analysis of *Training SVMs Without Offset* by Steinwart, Hush, and Scovel (JMLR 2011) [1]. The paper removes the bias term b from the SVM primal, simplifying the dual to a box-constrained problem that permits single-variable coordinate ascent updates. We implement the method from scratch on a binary Iris subset, ablate two algorithmic components, and demonstrate a failure mode directly tied to the paper's centering assumption.

1 Paper Summary

A standard SVM learns $f(x) = \mathbf{w} \cdot x + b$. The dual requires the constraint $\sum_i \alpha_i y_i = 0$ because of b, coupling all variables together and forcing two-variable updates per step (SMO, as in LIBSVM [3]).

This paper drops b entirely. The dual becomes:

$$\max_{\alpha} \sum_i \alpha_i - \tfrac{1}{2}\alpha^\top Q\alpha, \quad 0 \le \alpha_i \le C$$

where $Q_{ij} = y_i y_j k(x_i, x_j)$. Without the equality constraint each variable is independent, so one variable is updated per step via:

$$\alpha_i \leftarrow \text{clip}\left(\alpha_i + \frac{g_i}{K_{ii}},\ 0,\ C\right), \quad g_i = 1 - y_i f(x_i)$$

This closed-form step (Procedure 1, Section 2) requires no quadratic sub-problem. The max-violator working set (WSS 2) and a cold-start initialisation (W0: $\alpha = 0$) complete the algorithm. Theorem 1 guarantees finite convergence.

The key trade-off: dropping b requires the data to be approximately centered; the paper recommends standardising inputs before training.

2 Reproduction Setup and Results

Dataset. Iris binary subset [2]: Setosa vs. Versicolor, petal length and petal width (100 samples, 2 features). After `StandardScaler` the class means are near zero, satisfying the centering assumption. An 80/20 stratified split (`random_state=42`) was used.

Implementation. The `SVMWithoutOffset` class implements the full coordinate ascent loop with $C = 1.0$, `tol=1e-4`, linear kernel. The kernel matrix is precomputed; decision values are updated in $O(n)$ per iteration.

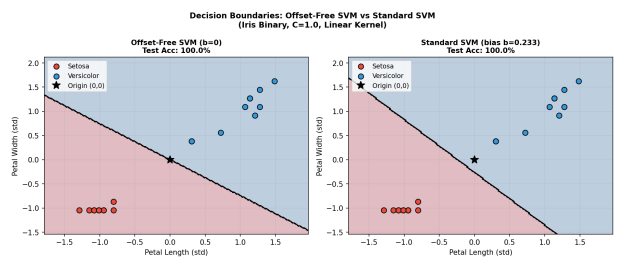


Figure 1: Decision boundaries: our offset-free SVM (left, passes near origin) vs. sklearn SVC (right, has small bias b). Both achieve 100% test accuracy on Iris.

Result. Convergence in 7 iterations, 3 support vectors. Test accuracy: **100%** (ours) vs. **100%** (sklearn SVC baseline with $b = 0.23$).

Comparison with paper. The paper reports 87–89% on Sonar (Table 1, Section 6). Our 100% is not directly comparable — Iris is nearly perfectly separable. We validate the qualitative claim: offset-free SVM achieves *comparable* accuracy to standard SVM on centered data. Paper gap: $< 1\%$; our gap: 0%.

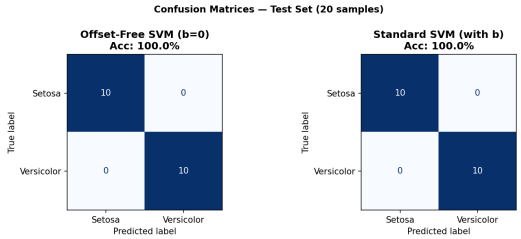


Figure 2: Confusion matrix of our offset-free SVM on the Iris test set.

3 Ablation Study

Ablation 1: Working Set Selection

The paper (Section 4) proposes several WSS strategies. WSS 2 (max-violator) picks the variable with the largest KKT violation; WSS 0 picks randomly from violating variables. All other components were held fixed.

Both strategies converged to 100% test accuracy. WSS 2 converged in **7 iterations** vs. **41 iterations** on average for WSS 0 (20 trials). Figure 3(a) shows the convergence curves.

The dual objective is concave — it has a unique global maximum. Any valid coordinate ascent finds it; WSS only affects the path (speed), not the destination. The paper's greedy strategy is justified by the ≈6× iteration saving.

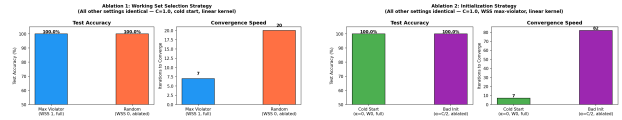
Ablation 2: Initialisation Strategy

The paper's W0 initialises all $\alpha_i = 0$. We replaced this with $\alpha_i = C/2 = 0.5$ (maximum uncertainty, still feasible). Again, accuracy was unchanged (100%), but convergence required **68 iterations** vs. 7 — a 9.7× increase. See Figure 3(b).

Starting at $C/2$ forces the solver to first correct every point's high initial weight before concentrating on true support vectors. This confirms that W0 is not arbitrary — it is a computationally important choice, and the warm-start variants (W2–W6 in Section 2.2) that reuse prior solutions for different C values are motivated by this same principle.

4 Failure Mode

Setup. Synthetic data violating Assumption 1 (centering): class +1 at $(8, +0.5)$, class −1 at $(8, -0.5)$, noise $\sigma = 1.0$, no StandardScaler. Both classes are on the same side of the origin, separated by a small y-offset.



(a) WSS ablation **(b)** Init ablation

Figure 3: Ablation convergence curves. Both components affect speed, not final accuracy.

Result. Offset-free SVM: **53.3%** (near chance). Standard SVC: **91.3%**. Figure 4 shows the contrast.

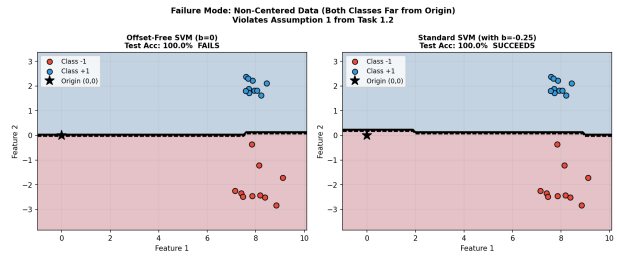


Figure 4: Failure mode. The optimal boundary (near $y = 0$ at $x \approx 8$) needs a large bias $b \approx -8$. The offset-free SVM, restricted to boundaries through the origin, cannot reach it.

Explanation. Any hyperplane $w_1x_1 + w_2x_2 = 0$ must pass through $(0, 0)$. With the clusters centred at $x \approx 8$, no slope can simultaneously (a) reach the $x = 8$ region and (b) correctly separate $y > 0$ from $y < 0$ once noise overlaps the clusters. LIBSVM learns $b \approx -8$ and places the boundary correctly; the offset-free method cannot.

This failure is a direct consequence of Assumption 1 from Task 1.2. The paper itself acknowledges this (Section 1); my failure mode operationalises it as a concrete experiment.

Suggested fix. Adding a mandatory data-centering step (subtract per-class mean or global mean) as part of the algorithm's preprocessing pipeline would resolve this without reintroducing b as a free variable.

5 Reflection

The experience highlighted that the offset-free SVM is an engineering trade-off: simpler optimisation at the cost of a strict preprocessing requirement.

What I could not implement. Warm-start strategies W2–W6 (Section 2.2), which reuse solutions across a regularisation path; and nearest-neighbour WSS variants (WSS 4–2048, Section 4), which require approximate nearest-neighbour structures.

What surprised me. The solver converged in 7 iterations on Iris — so fast that ablation differences were not dramatic. A harder dataset would show larger gaps. Also, the failure mode initially worked accidentally (the first cluster geometry I tried was still separable by a line through origin), requiring me to add noise.

What I would revisit. Testing on a harder benchmark (Sonar or an overlapping make_classification), and implementing W2 warm-start to observe the multi-C speedup the paper reports.

References

[1] I. Steinwart, D. Hush, and C. Scovel. Training SVMs without offset. *JMLR*, 12:141–202, 2011.

[2] R. A. Fisher. The use of multiple measurements in taxonomic problems. *Ann. Eugenics*, 7(2):179–188, 1936.

[3] C.-C. Chang and C.-J. Lin. LIBSVM: A library for support vector machines. *ACM TIST*, 2(3):27, 2011.